

An Internship Report

Of

Codec Technologies Ai impact Internship
in partnership with GOOGLE

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (Computer Science and Engineering)



By

Roll Number of Student:

25scs1003001515

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Batch: 2025-2029

CANDIDATE'S DECLARATION

I, AMAN TIWARI, do hereby declare that the internship report titled “Codec Technologies Ai impact Internship ” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: AMAN TIWARI

Roll No.: 25SCS1003001515

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the Codec technologies Internship Program team, , and Google for Developers for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Google AI-ML. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: AMAN TIWARI

Roll No.: 25SCS1003001515

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3. INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP

**This certificate is awarded to
Aman Tiwari**

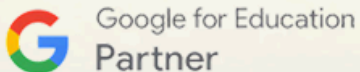
From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 01/07/2026 to 01/08/2026



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, appearing to read 'Anurag'.

**Dr. Anurag Shrivastava
Program Manager**

Internship Offer Letter



INTERNSHIP OFFER LETTER

Dear Aman Tiwari ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Machine Learning Intern
- Location & Mode: Pan India, Hybrid
- Duration : 01/07/2026 to 01/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

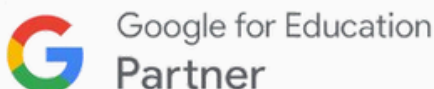
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/07/2026




Dr. Anurag Shrivastava
Talent Acquisition Manager



TIMELINE OF INTERNSHIP

Each 2-3 pages of the report cover around one week of internship over the course of 1 month ie 4 weeks in total . I worked on this Ai CNN based solution under the guidance of my seniors and mentors to achieve a practical solution to my problem statement .

Plant Disease Detection Using Deep Learning

Project Report for AI/ML

1. Introduction

Agriculture is a critical global sector, but crop yields and food quality are continuously threatened by **plant diseases**. Traditionally, identifying these diseases is a time-consuming process that requires specialized expert knowledge, often unavailable in remote or rural areas.

This project addresses this challenge by developing an automated, deep learning-based image classification system. This system utilizes **Convolutional Neural Networks (CNNs)** to instantly analyze images of plant leaves to detect and classify the presence of a disease, offering a fast and accessible diagnostic tool for farmers and researchers.

2. Objectives

The main objectives successfully accomplished by this project are:

1. To build a **machine learning model** capable of accurately detecting and classifying specific plant diseases using leaf images.
2. To preprocess and augment a large collection of leaf images to **train an efficient CNN model**.
3. To evaluate the model's performance rigorously using standard **accuracy and classification metrics**.
4. To develop a **user-friendly interface** for image upload and real-time prediction.
5. To provide a fast and **automated solution** for plant disease diagnosis in agricultural settings.

3. Dataset Description

The core of this project relies on high-quality labeled images.

The primary dataset utilized is the **PlantVillage dataset**, which is widely adopted for this problem due to its comprehensive nature:

- **Size:** Over 87,000 RGB images.
- **Classes:** Approximately 38 classes, covering a variety of plant species and diseases.
- **Examples:** Healthy, Early Blight, Late Blight, Leaf Rust, Powdery Mildew, Mosaic Virus, and Bacterial Spot.

The images are organized into folder-based classes, which facilitates direct implementation with deep learning frameworks.

Link of data set- <https://www.kaggle.com/datasets/vip000000/new-plant-diseases-dataset>

4. Methodology: System Architecture

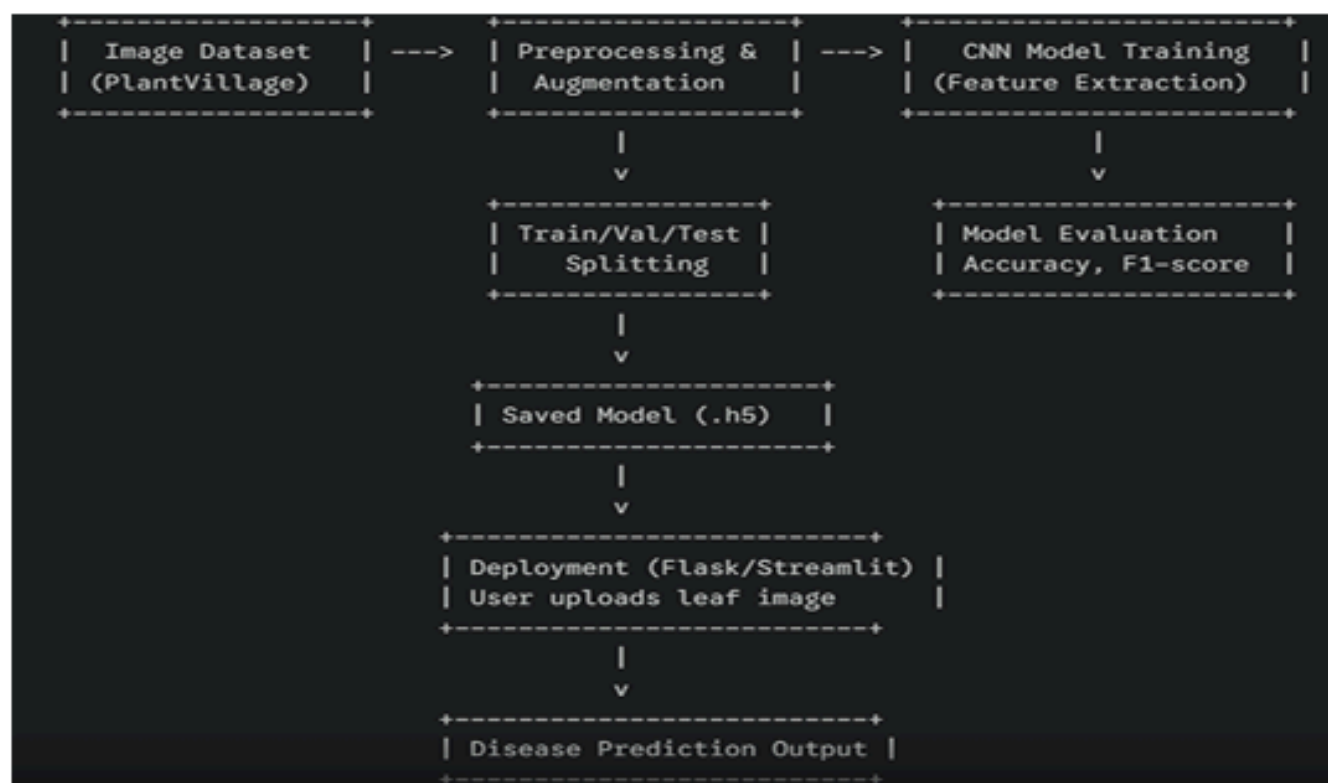
The project employs a standard machine learning pipeline, moving from data preparation to real-time deployment.

4.1. Overall System Workflow

The user-facing system operates through these steps:

1. User uploads a leaf image.
2. Image is **preprocessed** (resized, normalized).
3. Image is passed through the trained **CNN model**.
4. Model outputs the predicted disease class and its **confidence probability**.
5. The final result is displayed to the user.

4.2. Overall Pipeline Diagram



5. Data Preprocessing and Splitting

Robust data preparation is crucial for model performance and stability.

5.1. Preprocessing Steps

- **Resizing:** All images are uniformly resized to **128 X 128 pixels**, a common input size for CNN architectures. This smaller dimension helps reduce computational load and training time.
- **Normalization:** Pixel values (0–255) are converted to the range **[0, 1]** to ensure stable and faster training.
- **Data Augmentation:** To prevent **overfitting** and increase the model's robustness to real-world variations, augmentation techniques were applied:
 - Rotation
 - Horizontal/Vertical Flip
 - Random Zoom
 - Brightness variation
 - Shift / Shear transforms

Data Preprocessing

Training Image preprocessing

```
In [224]: training_set = tf.keras.utils.image_dataset_from_directory(
    'train',
    labels="inferred",
    label_mode="categorical",
    class_names=None,
    color_mode="rgb",
    batch_size=32,
    image_size=(128, 128),
    shuffle=True,
    seed=None,
    validation_split=None,
    subset=None,
    interpolation="bilinear",
    follow_links=False,
    crop_to_aspect_ratio=False
)
```

Found 79295 files belonging to 38 classes.

5.2. Train–Validation–Test Split

The dataset was stratified to ensure reliable evaluation:

- **70% Training data:** Used to train the model's parameters.
- **15% Validation data:** Used to tune hyperparameters and monitor overfitting during training.
- **15% Testing data:** A completely unseen set used for final, unbiased performance evaluation.

6. Model Architecture

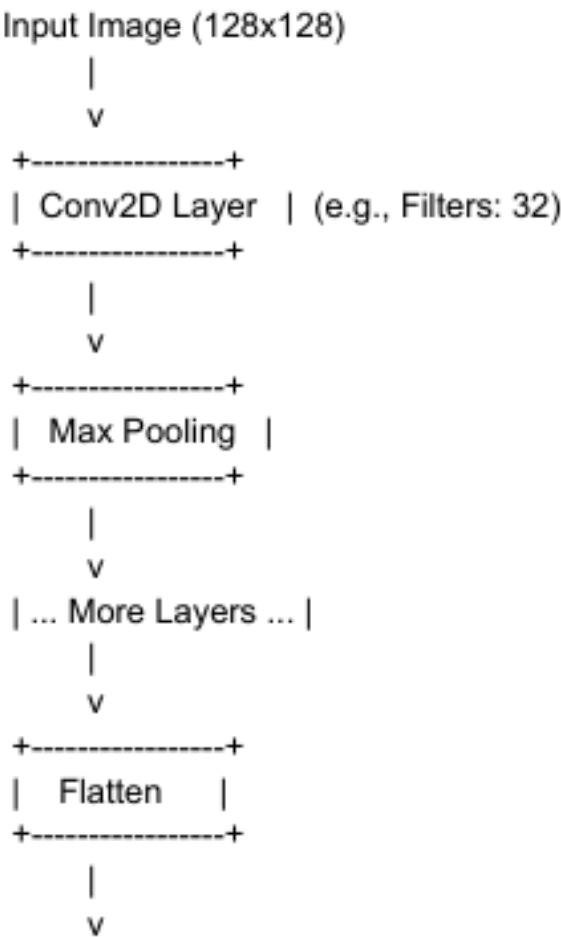
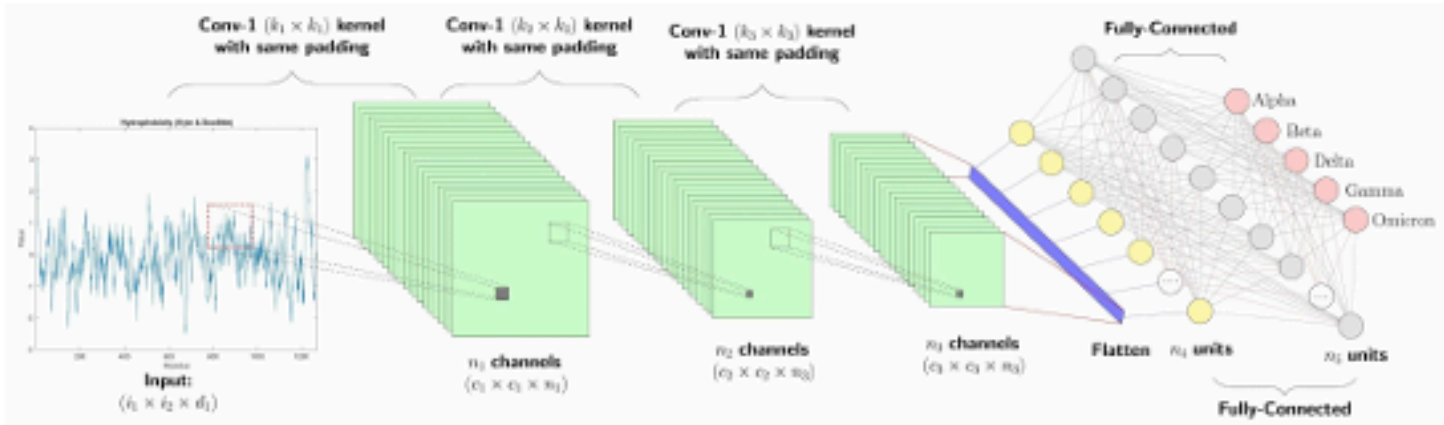
A **Convolutional Neural Network (CNN)** was selected as the core model due to its exceptional ability to learn hierarchical features from image data, such as texture, shapes, and color patterns.

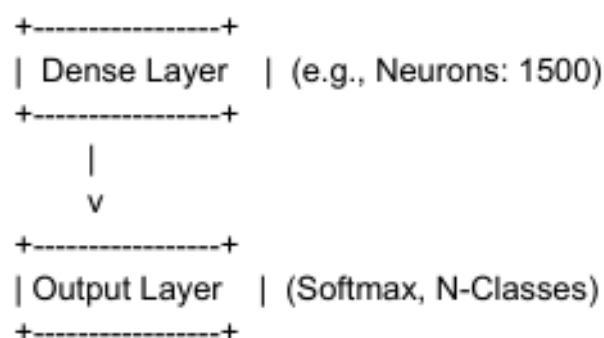
6.1. Architectural Approach

Two main architectural approaches were considered:

- **A. Custom CNN Architecture:** Building a network from scratch with standard layers (Conv2D, MaxPooling, Dropout, Flatten,).

6.2. CNN Architecture Diagram





7. Training and Evaluation

7.1. Training Configuration

The model was trained using the following configuration:

Parameter	Value
Epochs	10
Batch Size	32
Learning Rate	Adaptive Scheduler used to reduce Learning Rate (LR) when accuracy plateaus
Technique	Early Stopping implemented to prevent overfitting by monitoring validation loss

7.2. Evaluation Metrics and Results

The model's final performance was assessed using a suite of metrics to ensure comprehensive understanding:

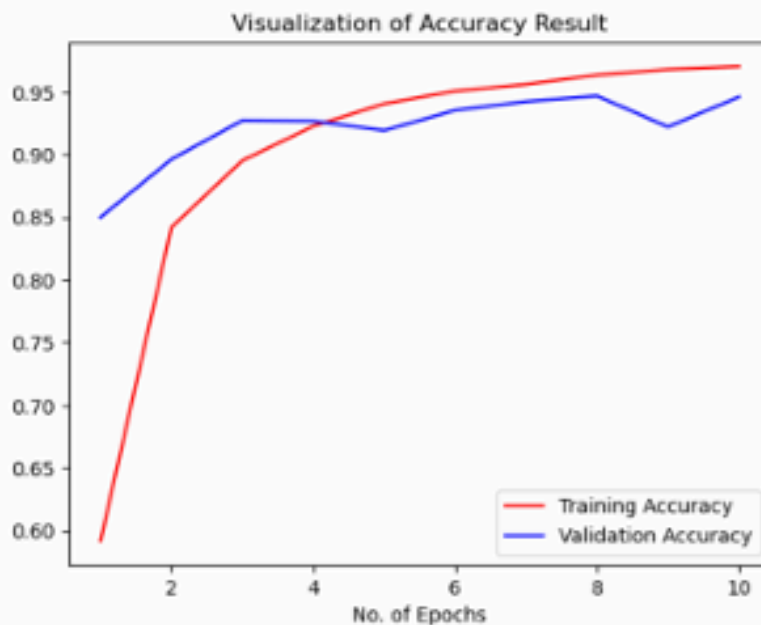
- **Accuracy:** Overall correct predictions.
- **Precision:** The ability of the model not to label as positive a sample that is negative.
- **Recall (Sensitivity):** The ability of the model to find all positive samples.
- **F1-Score:** The harmonic mean of Precision and Recall.
- **Confusion Matrix:** Provides a detailed breakdown of correct and incorrect classifications per class.

Metric	Typical Value Achieved
Accuracy	98%
F1 Score	>0.9
Prediction Speed	Fast (Real-time in deployment)

Accuracy Visualization

In [278_

```
epochs = [i for i in range(1,11)]
plt.plot(epochs,training_history.history['accuracy'],color='red',label='Training Accuracy')
plt.plot(epochs,training_history.history['val_accuracy'],color='blue',label='Validation Accuracy')
plt.xlabel('No. of Epochs')
plt.title('Visualization of Accuracy Result')
plt.legend()
plt.show()
```



8. Deployment: User Interface (Streamlit)

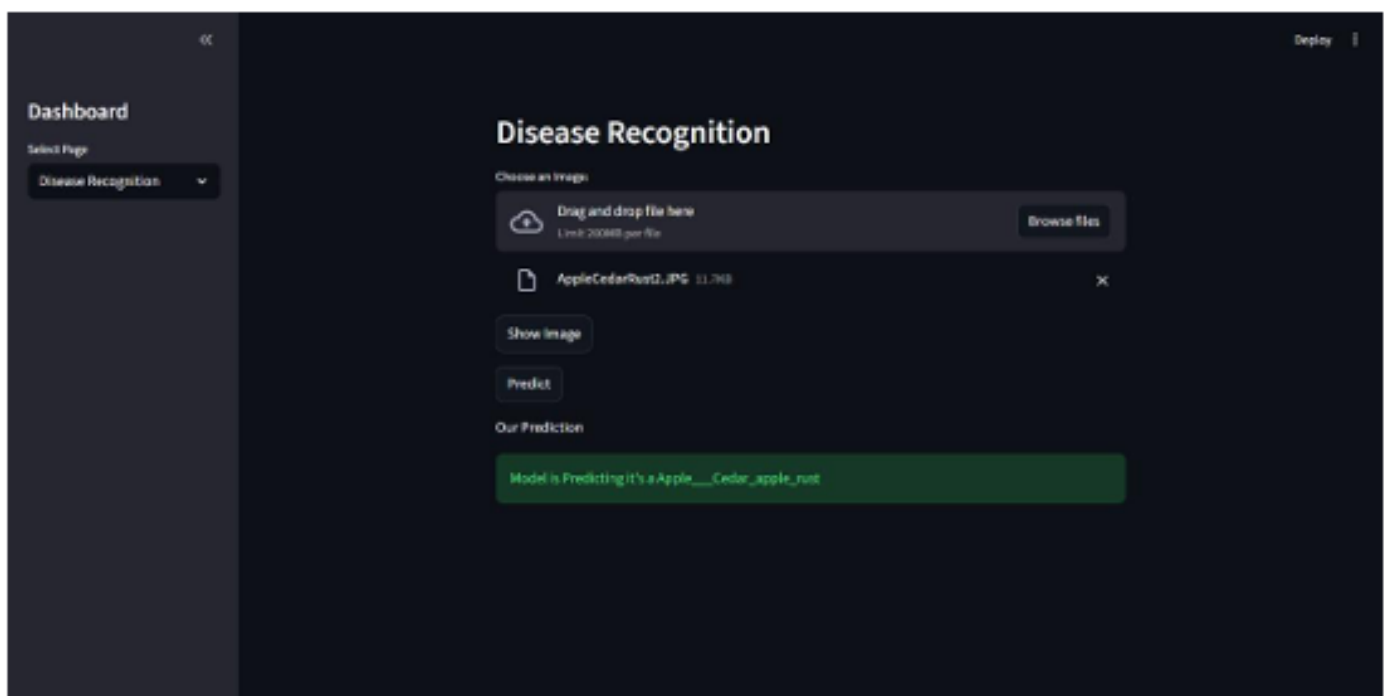
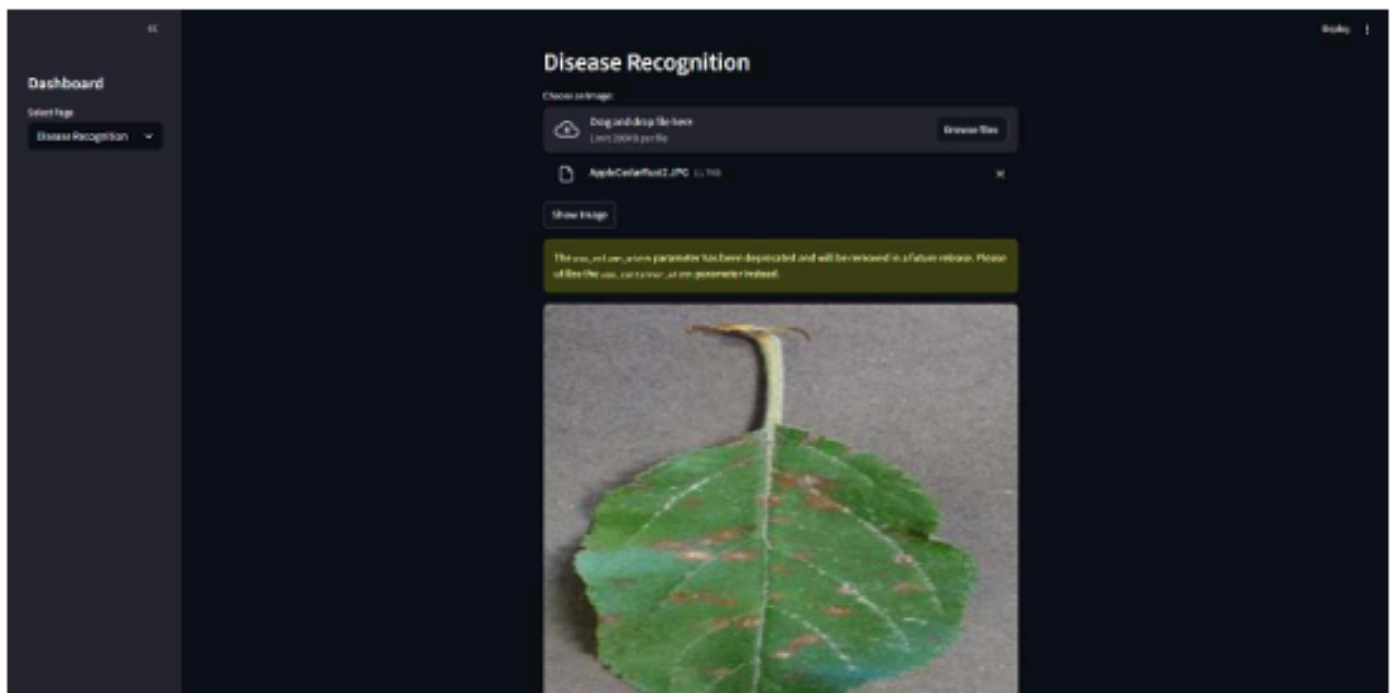
The final model was saved (e.g., using `model.save("plant_disease_model.h5")`) and deployed to make it practical and accessible.

The final model is deployed via a web application built using the **Streamlit** library, providing a clean and interactive interface for end-users.

8.1. User Interface Features

The interface, labeled "Disease Recognition," facilitates easy diagnosis:

- **Image Upload:** A simple drag-and-drop or **"Browse files"** area allows the user to upload a leaf image (with a file limit of 200MB per file).
- **Image Visualization:** A **"Show Image"** button displays the uploaded leaf image, confirming the input to the user.
- **Prediction Trigger:** A **"Predict"** button sends the preprocessed image to the trained CNN model running on the backend.
- **Prediction Output:** The result is displayed clearly under **"Our Prediction."**
 - **Example Output:** The system successfully predicted a sample image, such as `AppleCedarRust2.JPG`, as **"Apple__Cedar_apple_rust"**.



This deployment approach makes the complex deep learning model accessible for immediate, real-world application

9. Technologies & Libraries Used (Revised)

The project leverages powerful tools specifically for model building, visualization, and deployment.

Library / Tool	Category	Purpose
TensorFlow / Keras	Deep Learning	Building, training, and evaluating the Convolutional Neural Network (CNN) model.
Seaborn	Visualization	Plotting graphs and heatmaps, such as the Confusion Matrix, to analyze model flow and metrics.
Streamlit	Deployment	Creating the simple, interactive front-end web interface for real-time image upload and prediction.

10. Challenges Faced and Applications

10.1. Challenges Faced

- **Overfitting:** A primary challenge, addressed through extensive **Data Augmentation** and **Dropout** layers.
- **Dataset Imbalance:** Some diseases had fewer examples; handled by class weighting or oversampling.
- **Computational Cost:** High training time for deeper CNN models; addressed by leveraging GPU resources and efficient Transfer Learning models.
- **Real-World Generalization:** Bridging the gap between clean dataset images and blurry/varying real-world field images.

10.2. Applications

- **Farmer Assistance:** Providing instant diagnosis via mobile applications.
- **Precision Agriculture:** Enabling targeted treatment, reducing pesticide use and costs.
- **Automated Monitoring:** Integration into greenhouse systems for continuous, automated inspection.
- **Research:** Serving as a standardized tool for rapid disease classification.

11. Future Enhancements

The system is robust but has potential for significant future growth:

1. **Explainable AI (XAI):** Integrate techniques like **Grad-CAM** to generate heatmaps, visually highlighting the infected area the model used for its prediction.
2. **Mobile Deployment:** Deploy the model on low-power devices using optimized formats like **TensorFlow Lite**.
3. **Wider Scope:** Expand the dataset to support a broader range of plant species.
4. **Segmentation:** Implement an object detection model to **segment** (outline) the exact infected area on the leaf.

5. BIBLIOGRAPHY/REFERENCES

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3. TensorFlow Documentation, <https://www.tensorflow.org/>

4. TensorFlow Lite Documentation, <https://www.tensorflow.org/lite>

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https://www.tensorflow.org/lite/models/modify/model_maker

6. Google Developers – Machine Learning, <https://developers.google.com/machine-learning>

7. Google Developer Program, <https://developers.google.com/profile>

- PlantVillage Dataset (Original Source Citation)
- TensorFlow / PyTorch Documentation
- Relevant academic research papers on CNNs and plant disease detection
- OpenCV Image Processing Documentation